

Comparative Analysis of YOLO-based Deep Learning Models for Defect Detection in Product Packaging

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Abstract—In alignment with the goals of Industry 5.0, which emphasize intelligent, human-centric, and sustainable manufacturing, the integration of advanced quality control (QC) systems has become increasingly critical. This study presents a comprehensive comparative analysis of eight state-of-the-art YOLO based deep learning models, YOLOv3u, YOLOv5mu, YOLOv6m, YOLOv8m, YOLOv9m, YOLOv10m, YOLOv11m, and YOLOv12m for automated defect detection in product packaging. Using the Red Bull Can Appearance Defects Synthetic Dataset, each model was trained and tested under consistent conditions and assessed across key metrics mAP@0.5, mAP@0.5:0.95, precision, recall, F1-score, parameter count, FLOPs, frames per second (FPS), and latency. With the fastest inference (6.3 ms), highest FPS (158.7), and shortest training time (0.664 h), YOLOv5mu proved to be the ideal option for time-sensitive deployments. It also maintained high accuracy (mAP@0.5 = 0.950, F1 = 0.95). YOLOv9m had the best mAP@0.5:0.95 (0.599), precision (0.951), and a high F1-score (0.94) for tasks that required accuracy. The most balanced trade-off was provided by YOLOv10m, which combined effective resource use with high mAP@0.5 (0.952), F1-score (0.95), and low latency (7.0 ms). Despite YOLOv12m's incorporation of sophisticated modules like R-ELAN and FlashAttention, the current dataset did not significantly improve. This study provides a reference for the automatic recognition of defects in product packaging.

Keywords—yolo, defect detection, deep learning, industry 5.0, computer vision, quality control, model efficiency.

I. INTRODUCTION

The combination of robotics, machine learning, and artificial intelligence has led to developments in manufacturing automation. This has enabled it achievable to develop smart factories that can reduce human error, increase throughput, and cut down on material waste [1]. As

manufacturing transitions into Industry 5.0, quality control (QC) processes are transforming from reactive to proactive, smart solutions that combine AI capabilities with human insight to improve reliability, adaptability, and sustainability [2]. Traditional QC procedures that rely on manual inspection are becoming less and less practical, especially in high-speed production settings. Manual systems are labour-intensive, prone to human error, and frequently unable to identify minor or rapidly occurring flaws in real time [3], [4]. Furthermore, small-scale defects are another problem for traditional rule-based vision systems, which are frequently constrained by resolution limitations and insufficient processing speed, like those on semiconductor reels or complex packaging lines [5], [6]. AI-powered visual inspection systems have shown widespread use in industries like food processing and aerospace, where they have increased the precision and responsiveness of defect detection [7], [8]. However, there are still issues with system integration, scalability, and generalisation. Earlier machine learning-based approaches, including support vector machines (SVMs), have shown competent performance under controlled settings but lack the robustness needed in dynamic, real-world manufacturing conditions [9].

This study presents a comparative evaluation of eight YOLO-based architectures YOLOv3u, YOLOv5mu, YOLOv6m, YOLOv8m, YOLOv9m, YOLOv10m, YOLOv11m, and YOLOv12m focused on visual defect detection in product packaging to identify the best method in terms of accuracy, inference time, and computational efficiency. The scope of this study is limited to visual defects in product packaging and excludes non-visual defects such as weight inconsistencies or volume irregularities. Ultimately, this work contributes to the transition from traditional inspection systems to intelligent, sustainable, and